

Integration by Substitution

ANSWER KEY



u-substitution

- 1 Evaluate each integral using substitution:

a $\int 2x(x^2 + 1)^5 dx$ $u = x^2 + 1$: $= \frac{(x^2 + 1)^6}{6} + C$.

b $\int \frac{x}{\sqrt{x^2 + 9}} dx$ $u = x^2 + 9$: $= \sqrt{x^2 + 9} + C$.

c $\int \cos(3x) dx$ $u = 3x$: $= \frac{1}{3} \sin(3x) + C$.

- 2 Evaluate $\int x^2 e^{x^3} dx$.

$u = x^3$, $du = 3x^2 dx$: $= \frac{1}{3} e^{x^3} + C$.

Definite integrals with substitution

- 3 Evaluate $\int_0^2 x \sqrt{x^2 + 1} dx$ exactly, changing the limits with $u = x^2 + 1$.

u goes from 1 to 5: $\frac{1}{2} \int_1^5 \sqrt{u} du = \frac{1}{2} \left[\frac{2}{3} u^{3/2} \right]_1^5 = \frac{1}{3} (5\sqrt{5} - 1)$.

- 4 Evaluate $\int_0^{\pi/2} \sin^3 x \cos x dx$.

$u = \sin x$, limits 0 to 1: $\int_0^1 u^3 du = \frac{1}{4}$.

- 5 Evaluate $\int_1^e \frac{\ln x}{x} dx$.

$u = \ln x$, limits 0 to 1: $\int_0^1 u du = \frac{1}{2}$.